

ICP Server: Control Flow

The Identity Nervous System for Multi-Agent Enterprise Ecosystems

Core Mechanics

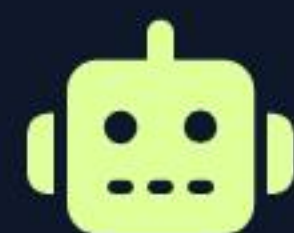
Defining the handshake between human authority, agent intent, and identity translation.

| Defined Persona Roles



Human Operator

The root identity. Authenticates via SSO (Ping/Entra) and provides the initial cryptographically signed OIDC token.



Primary Agent

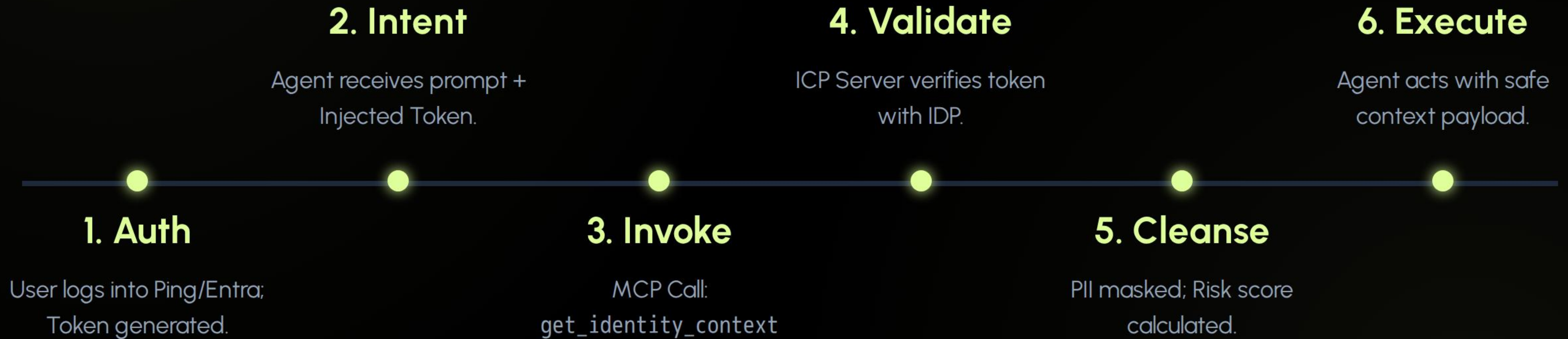
The MCP Host. Interacts with the human, captures the token, and acts as the secure "Token Firewall" layer.



Sub-Agent

The Delegated Worker. Performs specific background tasks under a traced provenance chain of authority.

| Standard Control Flow



| The Token Firewall Flow

Out-of-Band Security

The core security USP of the ICP Server is the **Agent Injection** model:

- 🛡️ **Isolation:** Tokens are handled by the host wrapper, not the LLM.
- 🚫 **Injection Defense:** Malicious prompts cannot leak tokens because the LLM context never sees them.
- ✅ **Clean Data:** Only sanitized booleans (e.g., `can_read: true`) enter the prompt.



| Enterprise IDP Connectivity



Microsoft Entra ID

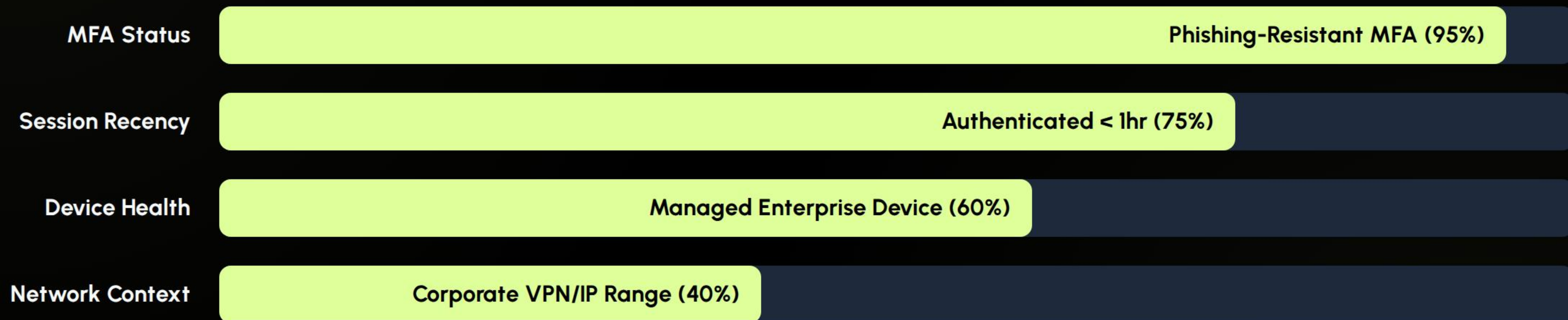
Native integration with Graph API for real-time group membership and Conditional Access telemetry.



Ping Identity

Support for PingOne and PingFederate flows, leveraging advanced risk signals from the PingProtect suite.

Risk Evaluation Logic



Risk Score (0-100) is calculated by weighting these real-time signals from the IDP response.

| Context Sanitization Table

Identity Signal	Raw Data (Unsafe)	ICP Process	Masked Context (Safe)
User Email	sarah.johnson@ge.com	SHA-256 Hashing	usr_8f93a1c2...
Access Groups	CN=Prod_DB_Admins,OU=...	Boolean Mapping	can_write_prod: true
IP Metadata	192.168.1.45 (Home)	Risk Evaluation	risk_level: "MEDIUM"

Multi-Hop Provenance

In complex enterprise workflows, agents often spawn sub-agents. The ICP Server tracks the entire **Identity Provenance Chain**:

-  **Traceability:** Human → Primary Agent → Sub-Agent.
-  **Auditability:** Logs exactly which human authorized the chain.
-  **Delegation:** Bounded authority passed down the stack safely.

| Enterprise Compliance Flow

-  **Zero-Log Token Policy:** Raw bearer tokens are never persisted or logged in the ICP lifecycle.
-  **TTL Caching:** Context is cached for < 5 mins to ensure high performance without stale security.
-  **Schema Validation:** Strict JSON-schema enforcement on all MCP tool inputs and outputs.
-  **Stateless Execution:** Designed for horizontal scaling in Kubernetes or Serverless environments.

| Performance Metrics

<200ms

Avg. Latency

Efficiency by Design

The Node.js/TypeScript core is optimized for sub-200ms response times, ensuring that adding an identity layer does not degrade the user experience of the AI Agent session.

By using in-memory TTL caching, subsequent tool calls within the same session are resolved in **<10ms**.

Questions?

Securing the future of Agentic Identity.

www.identity-context-protocol.io | developer@icp-server.io

| Image Sources



<https://en.pimg.jp/100/300/285/1/100300285.jpg>

Source: www.pixtastock.com



https://media.istockphoto.com/id/1665910516/photo/cloud-and-edge-computing-technology-data-transfer-concept-a-large-cloud-icon-is-in-the-center.jpg?s=612x612&w=0&k=20&c=v37ipYB5w6g47pk9bv1b0hcRJBuiyJ7X_WE2q8OEW8o=

Source: www.istockphoto.com